Transportation Data Visualization and Driving Behaviour Simulation Technical Report – Project #3

Client: Federal Highway Administration (FHWA)

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March 31, 2026

Table of Contents:

1. Introduction 2
2. Methods 3
2.1 Data sources and project workflow: 3
2.2 Variable selection and rationale 3
2.3 Data preparation and visualization: 3
2.4 Intelligent Driver Model simulation: 4
3. Results and Discussion 4
3.1 Household transportation patterns from NHTS 5
3.2 Additional fleet characteristics from the customized plot 6
3.3 Driving behaviour from NGSIM time-series plots 7
3.4 Simulation results and validation 9
4.Conclusion 9
5. References 10

1. Introduction

Transportation systems are essential components of modern society, supporting economic development and daily mobility. To improve transportation planning and decision-making, it is important to understand both travel behaviour and vehicle dynamics. The Federal Highway Administration (FHWA) has provided two datasets—the National Household Travel Survey (NHTS) and the Next Generation Simulation (NGSIM)—to support this analysis.

The NHTS dataset provides information on household-level travel behaviour, including vehicle ownership, demographics, and travel patterns. In contrast, the NGSIM dataset provides high-resolution vehicle trajectory data, such as position, speed, and acceleration, allowing for detailed analysis of driving behaviour and vehicle interactions.

The objective of this project is to analyse transportation trends and driving behaviour using these datasets. Specifically, this project aims to:

- Develop clear and meaningful visualizations of transportation data
- Identify key trends in household travel behaviour
- Analyse vehicle-following behaviour using trajectory data
- Apply a simulation model to better understand traffic flow dynamics

The results of this analysis provide valuable insights for transportation engineers and planners, helping improve roadway design, traffic management, and safety.

2. Methods

2.1 Data sources and project workflow:

Two datasets were used in the analysis. The NHTS file contained records and variables including household and vehicle characteristics such as household size, vehicles per household, household location, vehicle type, fuel type, vehicle age, and vehicle year. The NGSIM file contained time, leader position, follower position, leader speed, follower speed, leader acceleration, follower acceleration, and trajectory number. The client prompt and codebooks were reviewed before plotting so that the selected variables would be representative of transportation trends and driving behaviour.

2.2 Variable selection and rationale

Our team selected variables to match the rubric requirement that the report explain which variables were chosen and why they were meaningful. From NHTS, we choose vehicle_type for a categorical comparison because it shows what types of vehicles dominate the sample

and provides a simple view of household travel preferences. vehicles_per_household was selected for the histogram because it is a numerical variable directly related to access to private transportation. vehicle_age and household_location were selected for the boxplot because comparing urban and rural households provides a useful planning-oriented view of differences in vehicle replacement patterns and mobility context. fuel_type was selected for the customized plot because powertrain mix is closely related to long term transportation policy, fleet modernization, and energy transition (NHTS, n.d.).

From NGSIM, our team selected follower_speed(m/s) for a multi-trajectory time-series plot because speed over time directly reflects changes in traffic conditions and driver response. Gap distance, computed as leader_position(m) minus follower_position(m), was selected as the second time-series variable because spacing between vehicles is central to traffic safety and car-following behavior (Treiber et al.). For the simulation study, trajectory 1 was selected and the leader and follower acceleration records were compared against an IDM-simulated follower acceleration, which directly addressed the client's requirement for an acceleration comparison plot (Treiber et al.)

2.3 Data preparation and visualization:

The team used python to conduct analysis. The NHTS and NGSIM Excel files were loaded with pandas, and plotting was completed with matplotlib and seaborn (Matplotlib, n.d.; Seaborn, n.d.). We created the bar chart by counting the frequency of each vehicle_type category and plotting those counts with rotated x-axis labels to improve readability. Similarly, we created histogram from vehicles_per_household using integer bin widths so that discrete household vehicle counts were represented clearly. The boxplot was created by grouping vehicle_age according to household_location to compare the median, interquartile range, whiskers, and outliers for urban and rural households. The first time-series plot overlaid follower speed versus time for trajectories 1 through 4. The second time-series plot computed gap distance for trajectory 1 and plotted that spacing over time. We customized fuel-type bar chart to demonstrate plot customization.

2.4 Intelligent Driver Model simulation:

The simulation study used the Intelligent Driver Model (IDM), a car-following model that estimates follower acceleration based on current speed, desired speed, spacing to the leader, and relative speed (Treiber et al.). In this project, our team selected trajectory 1 from the NGSIM dataset for simulation, and the median time step was estimated from the time vector as approximately 0.100 seconds. The IDM parameters used were a desired speed of 30.0 m/s, minimum spacing of 2.0 m, desired time headway of 1.5 s, maximum acceleration of 1.0 m/s^2, comfortable deceleration of 1.5 m/s^2, and an acceleration delta (exponent) of 4. The simulated follower was initialized with the real follower's initial position and speed. At each

time step, the model computed the instantaneous gap, calculated the follower's relative speed to the leader, evaluated the IDM acceleration function, and updated follower speed and position. The final plot compared leader acceleration from NGSIM, real follower acceleration from NGSIM, and follower acceleration from the IDM simulation

3. Results and Discussion

3.1 Household transportation patterns from NHTS

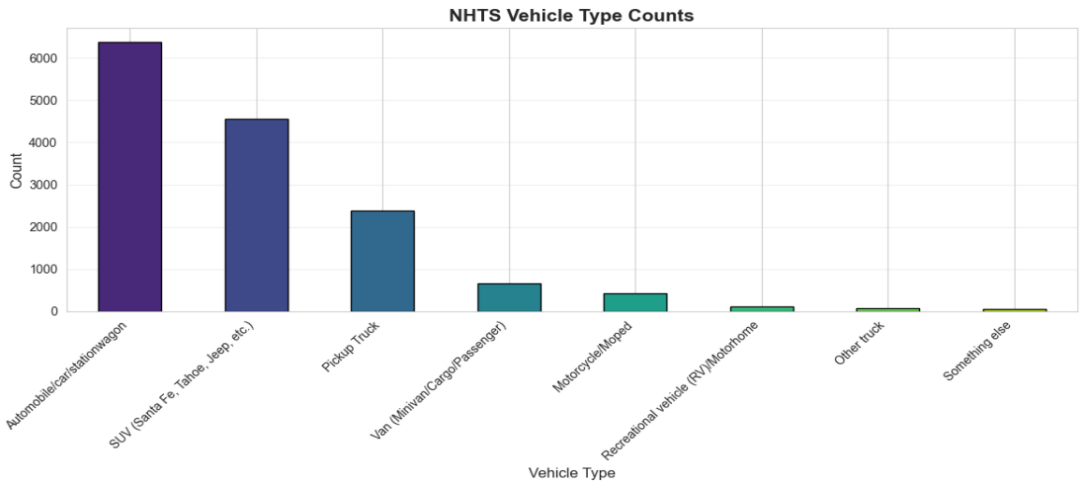

Figure 1 NHTS vehicle type counts.

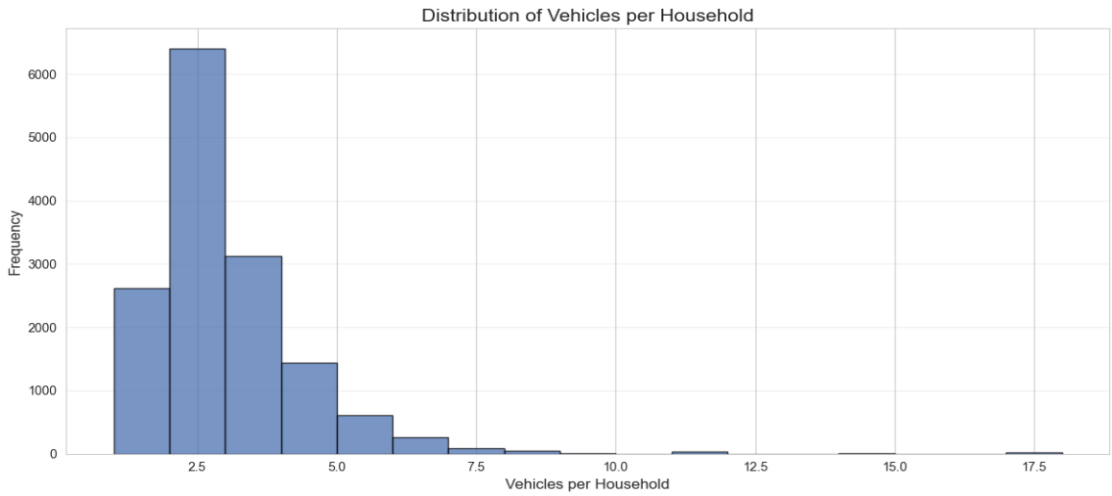

Figure 2 Distribution of vehicles per household.

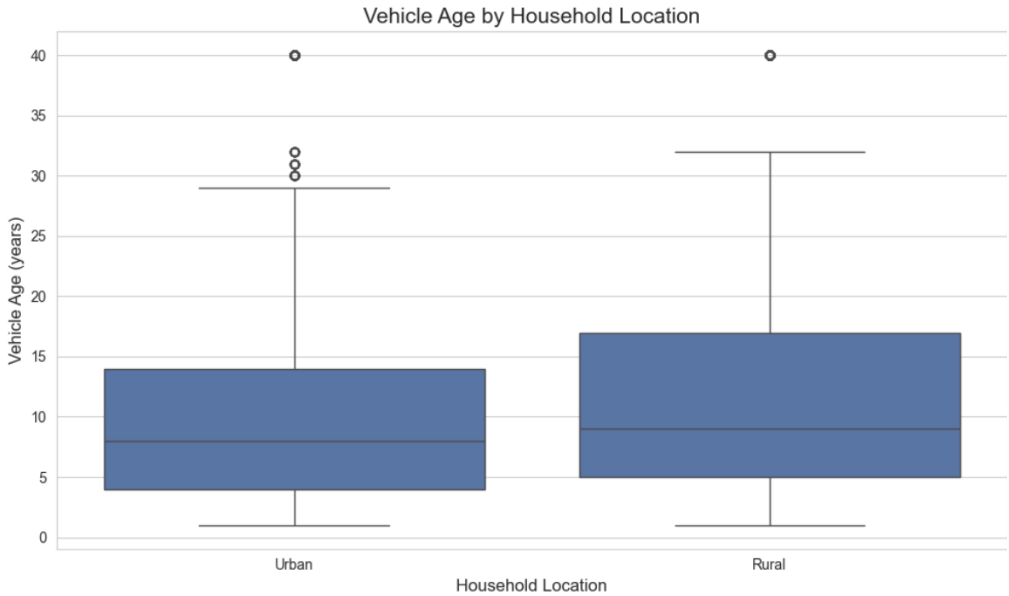

Figure 3 Vehicle age by household location.

Figure 1 shows that automobiles/cars/station wagons are the most common vehicle type in the NHTS sample, followed by SUVs and then pickup trucks. Vans, motorcycles/mopeds, recreational vehicles, and other truck categories appear much less frequently. This pattern suggests that the household fleet in the dataset is dominated by conventional light duty passenger vehicles. For the transportation industry, this matters because roadway design, parking demand, fuel use, and safety planning are all strongly influenced by the types of vehicles most used.

Figure 2 shows that the distribution of vehicles per household is concentrated around two to three vehicles, with the frequency dropping as vehicle ownership increases. The histogram has a right skewed shape, indicating that households with very large vehicle counts are uncommon but do exist in the dataset. From a planning perspective, this suggests that most households have regular access to private transportation, while a smaller number of households have unusually high vehicle availability.

Figure 3 compares vehicle age in urban and rural households. The median vehicle age appears slightly higher for rural households, and the rural distribution also spans a broader range. Both categories contain outliers, including very old vehicles near 40 years. The plot suggests that rural households may retain vehicles longer on average than urban households. This finding is relevant because older vehicles can be associated with different maintenance needs, fuel efficiency levels, and emission profiles. For the transportation industry, differences in fleet age between urban and rural areas may affect infrastructure demand, vehicle inspection programs, and the pace at which cleaner vehicle technologies penetrate different regions.

3.2 Additional fleet characteristics from the customized plot

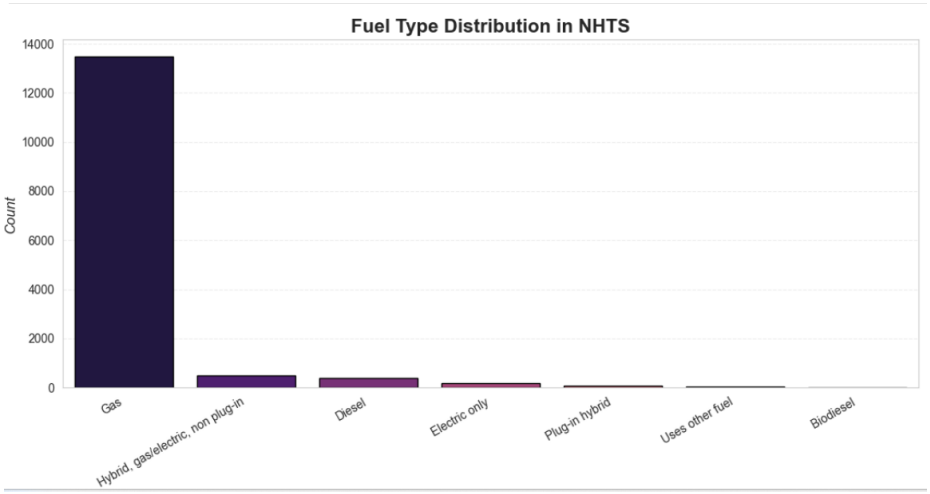

Figure 4 Fuel type distribution in NHTS.

Figure 4 shows the dominance of gasoline vehicles in NHTS sample. Gas vehicles have most numbers while Hybrid, Gas/electric/non-plug-in, Diesel and other fuel categories are present in small amount. This reflects the real-world reality that gasoline remains by far the most common vehicle fuel type in the dataset. Even though alternative fuel vehicles are increasing, they are not yet widely used compared to traditional vehicles. This is important for transportation planning because it shows that fuel infrastructure for gasoline is still necessary, while also highlighting the need to slowly expand support for electric and hybrid vehicles in the future.

3.3 Driving behaviour from NGSIM time-series plots

Figure 5 Follower speed versus time across multiple trajectories.

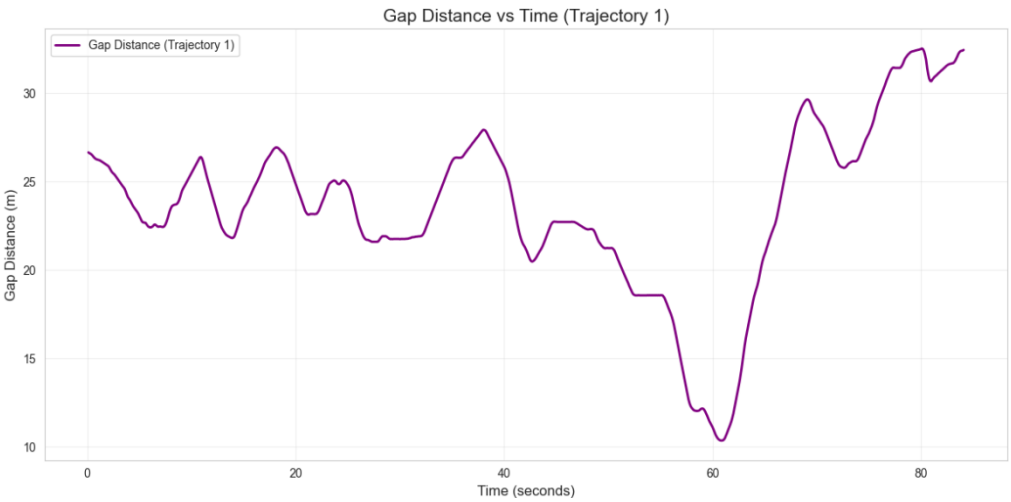

Figure 6 Gap distance versus time for trajectory 1.

Fig 5 shows follower speed over time for four different trajectories (NGSIM, n.d.). The results show that vehicle speed is not constant and changes frequently as time progresses. In some parts of the graph, the speed decreases significantly, which likely represents slowing down due to traffic or congestion. In other parts, the speed increases again, showing that vehicles accelerate when traffic conditions improve. Each trajectory follows a slightly different pattern, which shows that driving behaviour can vary depending on traffic conditions and driver responses. This figure demonstrates how drivers continuously adjust their speed based on surrounding vehicles. This is important for transportation engineering because it helps explain how traffic flow changes over time and how congestion can develop on roadways.

Fig 6 shows the gap distance over time for Trajectory 1. The gap fluctuates throughout but remains positive, narrowing to around 10 to11 meters during the most congested period before increasing again toward the end. This changing spacing reflects how the driver responds to traffic conditions ahead, and understanding these gap dynamics is useful for traffic simulation, and safety analysis. Smaller gaps can increase the risk of collisions, while larger gaps can reduce roadway efficiency. Understanding how gap distance changes over time helps engineers design safer and more efficient transportation systems.

3.4 Simulation results and validation

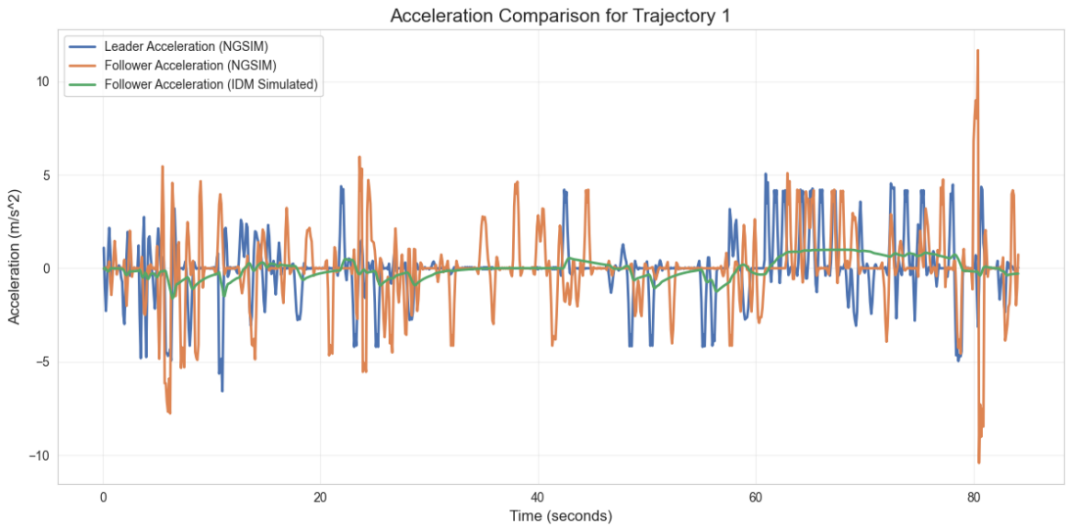

Figure 7 Acceleration comparison for trajectory 1.

Fig 7 compares leader acceleration, real follower acceleration, and IDM-simulated follower acceleration for trajectory 1. The graph shows that acceleration changes frequently, with both positive values (speeding up) and negative values (slowing down). The real follower closely follows the leader changes, showing how drivers react to the vehicle ahead. The simulated acceleration using the IDM model is smoother compared to the real data, but it still follows the overall trend. This shows that the model can capture general driving behaviour is important because acceleration behaviour help engineers understand how drivers respond to traffic conditions and evaluate how well simulation models represent real traffic flow.

4.Conclusion

This project analysed transportation behaviour and driving patterns using the NHTS and NGSIM datasets. The analysis by our team using NHTS dataset showed that the sampled fleet is dominated by automobiles, SUVs, and pickup trucks, that most households own approximately two to three vehicles, the rural households appear to operate somewhat older vehicles than urban households, and that gasoline remains the dominant fuel type by a wide margin. Using the NGSIM dataset, the analysis showed that follower speed and gap distance vary substantially over time, reflecting the dynamic nature of real traffic flow (Treiber et al.)

Our team's analysis on the IDM simulation for trajectory 1 shows that it produced a follower acceleration profile that was smoother than the measured NGSIM acceleration data but still captured the main behavioural trends (Kesting et al.). This indicates that the IDM can provide a practical approximation of real car-following behaviour for transportation simulation studies.

To sum up this project highlights the importance of data analysis and simulation in transportation engineering. The findings can help engineers better understand traffic behaviours, improve roadway design, and support future transportation planning.

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